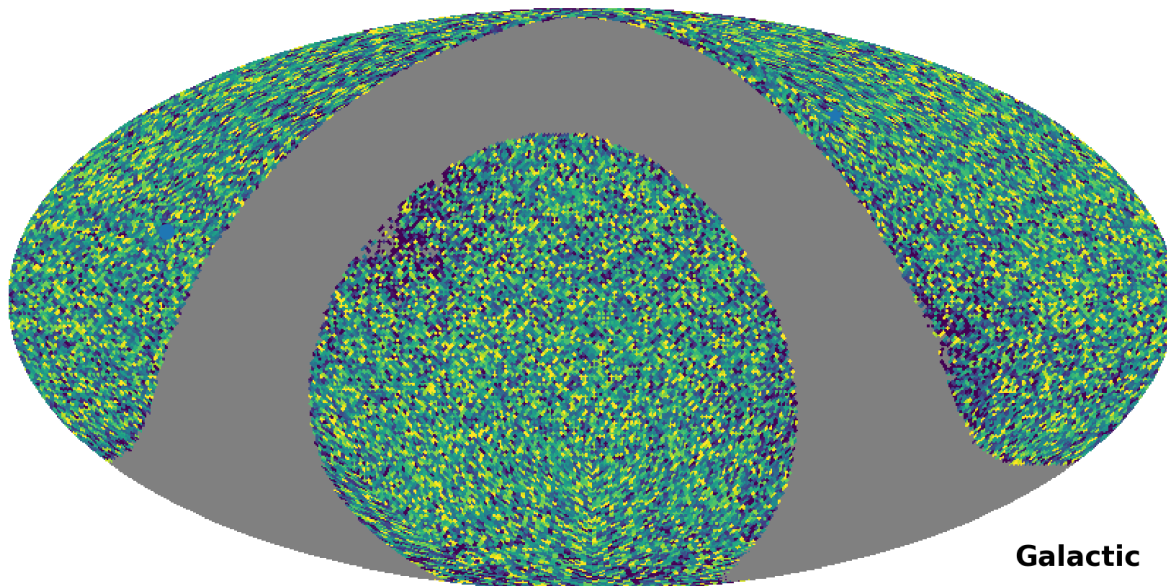


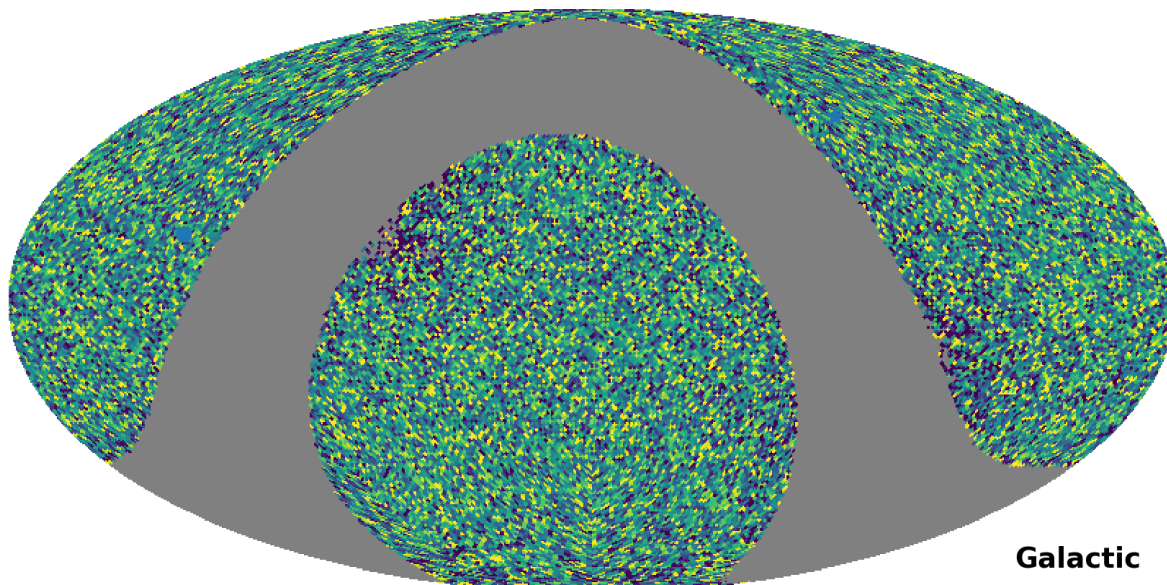
| #       | tag        | label      | N_good        | amp    | l_deg | b_deg | a0 | sep_cmb                  |
|---------|------------|------------|---------------|--------|-------|-------|----|--------------------------|
|         | full       |            | full sample   |        |       |       |    | 32388 8.9620e-03 110.947 |
| 18.511  |            | 5.9903e-03 | 109.25        |        |       |       |    |                          |
|         | z0p10_0p30 | 0.10       | ,â\$ z < 0.30 |        |       |       |    | 209 9.0275e-03           |
| 325.482 |            | 51.125     | 4.4684e-04    | 38.83  |       |       |    |                          |
|         | z0p30_0p50 | 0.30       | ,â\$ z < 0.50 |        |       |       |    | 3237 1.8290e-03          |
| 81.822  |            | -17.451    | 7.3655e-05    | 149.40 |       |       |    |                          |
|         | z0p50_0p75 | 0.50       | ,â\$ z < 0.75 |        |       |       |    | 7675 1.5889e-03          |
| 155.871 |            | -17.471    | -9.9769e-05   | 114.94 |       |       |    |                          |
|         | z0p75_1p00 | 0.75       | ,â\$ z < 1.00 |        |       |       |    | 12687 1.3108e-03         |
| 204.712 |            | -45.303    | 5.4196e-05    | 106.73 |       |       |    |                          |
|         | z1p00_1p50 | 1.00       | ,â\$ z < 1.50 |        |       |       |    | 28058 1.6622e-03         |
| 34.871  |            | 12.299     | 9.7139e-05    | 105.64 |       |       |    |                          |
|         | z1p50_2p50 | 1.50       | ,â\$ z < 2.50 |        |       |       |    | 30800 2.9533e-03         |
| 91.779  |            | -9.949     | 2.6834e-04    | 141.39 |       |       |    |                          |

full sample ( $|b| > 20^\circ$ )  
 $\langle z \rangle$  dipole: amp= $8.11\text{e-}03$ ,  $(l,b)=(134.8^\circ, 17.1^\circ)$



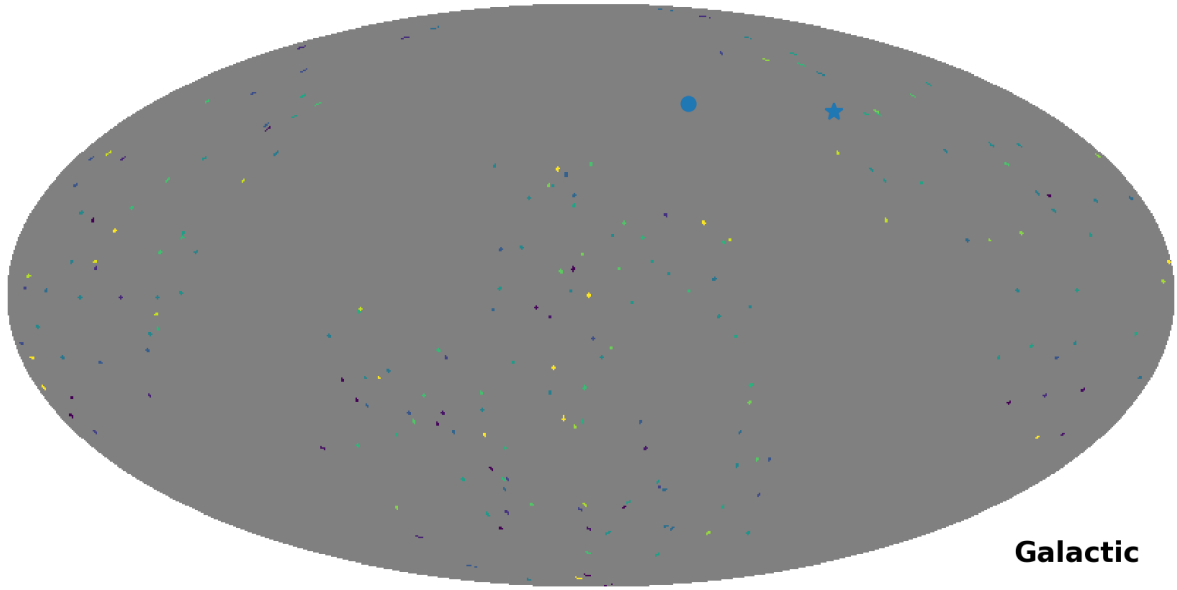
1.22909  $\langle z \rangle$  1.79662

full sample ( $|b| > 20^\circ$ ) residual  
 $z_{\text{resid}} = z - (a + bN)$ ,  $a=1.447$ ,  $b=3.38\text{e-}03$   
dipole: amp= $4.40\text{e-}03$ ,  $(l,b)=(129.1^\circ, 16.6^\circ)$



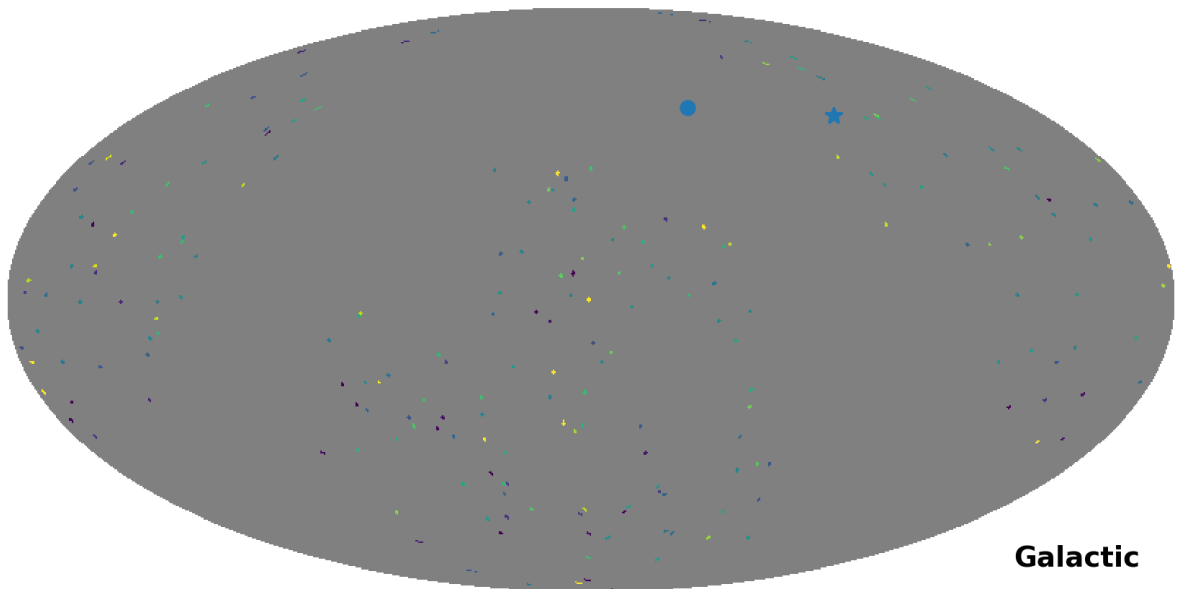
-0.277686  $z_{\text{resid}}$  0.284283

$0.10 \leq z < 0.30$  ( $|b| > 20^\circ$ )  
 $\langle z \rangle$  dipole: amp= $9.02 \times 10^{-3}$ ,  $(l, b) = (320.4^\circ, 50.5^\circ)$



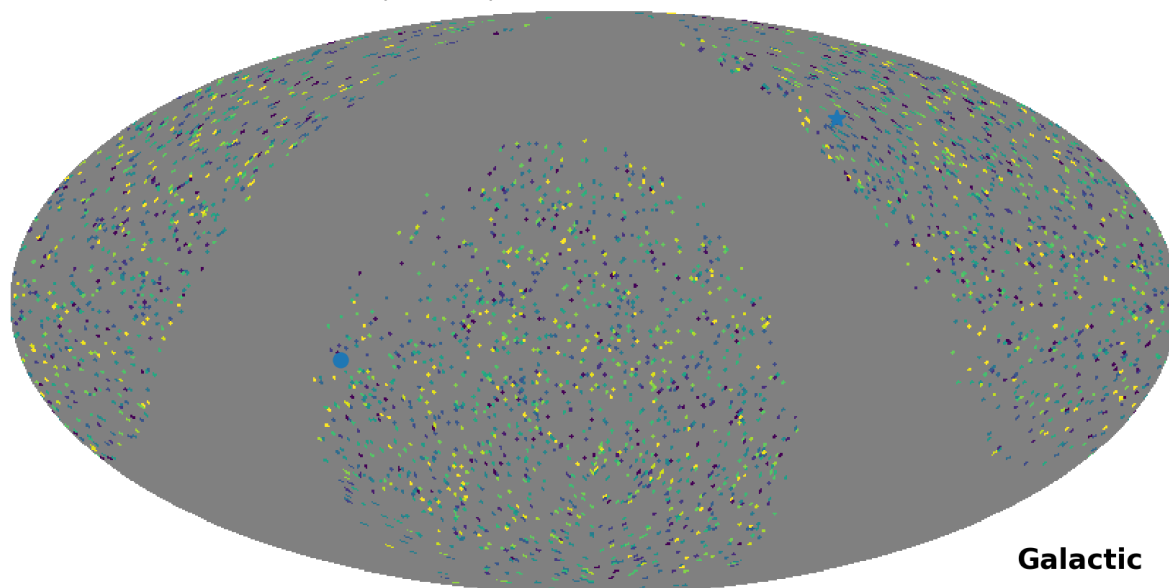
**Galactic**

$0.10 \leq z < 0.30$  ( $|b| > 20^\circ$ ) residual  
 $z_{\text{resid}} = z - (a + bN)$ ,  $a = 0.240$ ,  $b = -1.52 \times 10^{-3}$   
dipole: amp= $8.92 \times 10^{-3}$ ,  $(l, b) = (320.7^\circ, 50.2^\circ)$

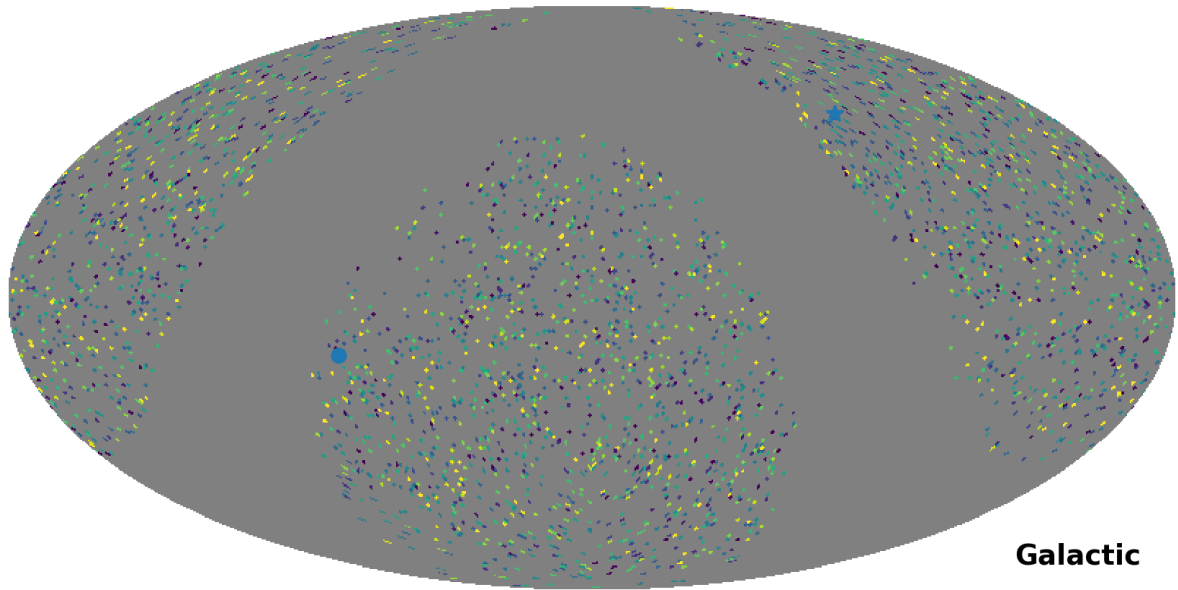


**Galactic**

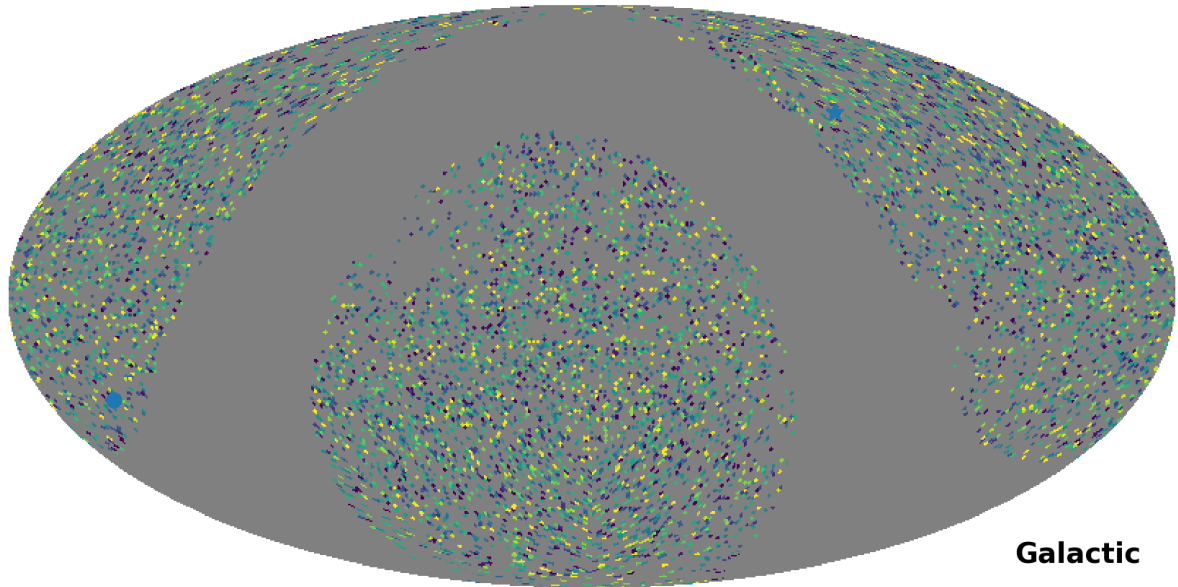
$0.30 \leq z < 0.50$  ( $|b| > 20^\circ$ )  
<z> dipole: amp=1.85e-03, (l,b)=(79.5°, -14.4°)



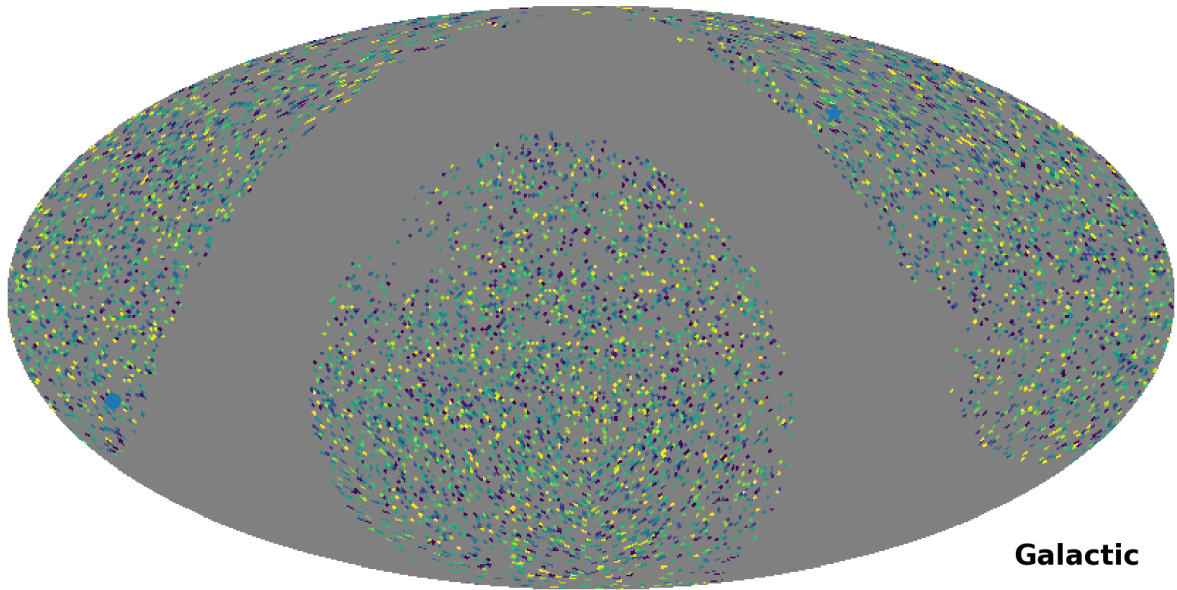
$0.30 \leq z < 0.50$  ( $|b| > 20^\circ$ ) residual  
 $z_{\text{resid}} = z - (a + bN)$ ,  $a=0.404$ ,  $b=6.92\text{e-}05$   
dipole:  $\text{amp}=1.85\text{e-}03$ ,  $(l,b)=(79.5^\circ, -14.4^\circ)$



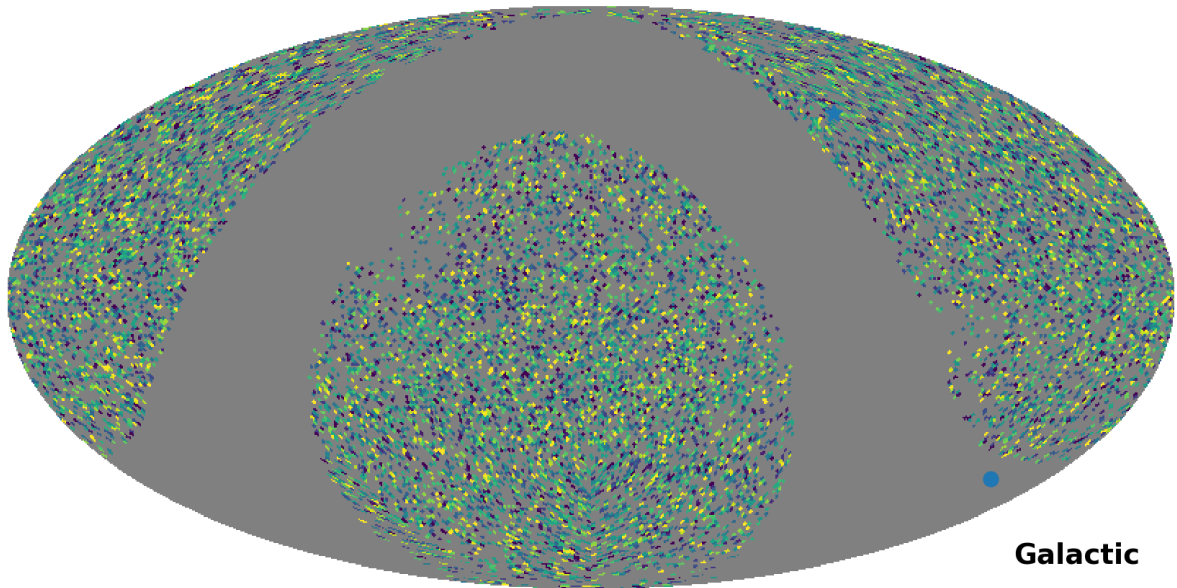
$0.50 \leq z < 0.75$  ( $|b| > 20^\circ$ )  
 $\langle z \rangle$  dipole:  $\text{amp}=1.40\text{e-}03$ ,  $(l,b)=(157.4^\circ, -26.1^\circ)$



$0.50 \leq z < 0.75$  ( $|b| > 20^\circ$ ) residual  
 $z_{\text{resid}} = z - (a + bN)$ ,  $a=0.631$ ,  $b=-2.61\text{e-}04$   
 dipole:  $\text{amp}=1.41\text{e-}03$ ,  $(l,b)=(157.4^\circ, -26.1^\circ)$

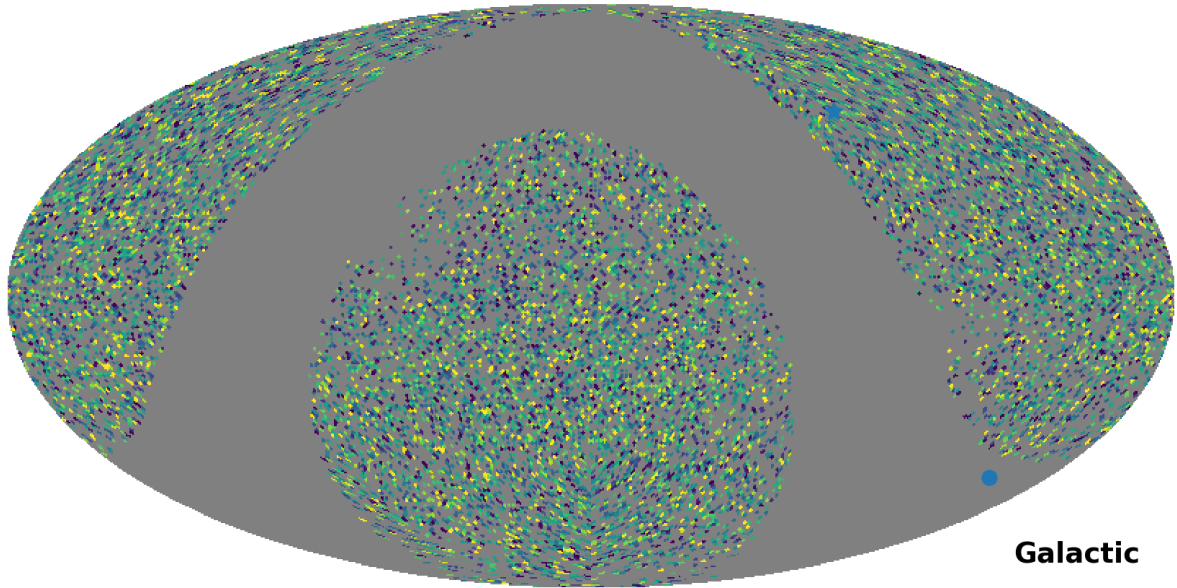


$0.75 \leq z < 1.00$  ( $|b| > 20^\circ$ )  
 $\langle z \rangle$  dipole:  $\text{amp}=1.61\text{e-}03$ ,  $(l,b)=(203.1^\circ, -47.5^\circ)$

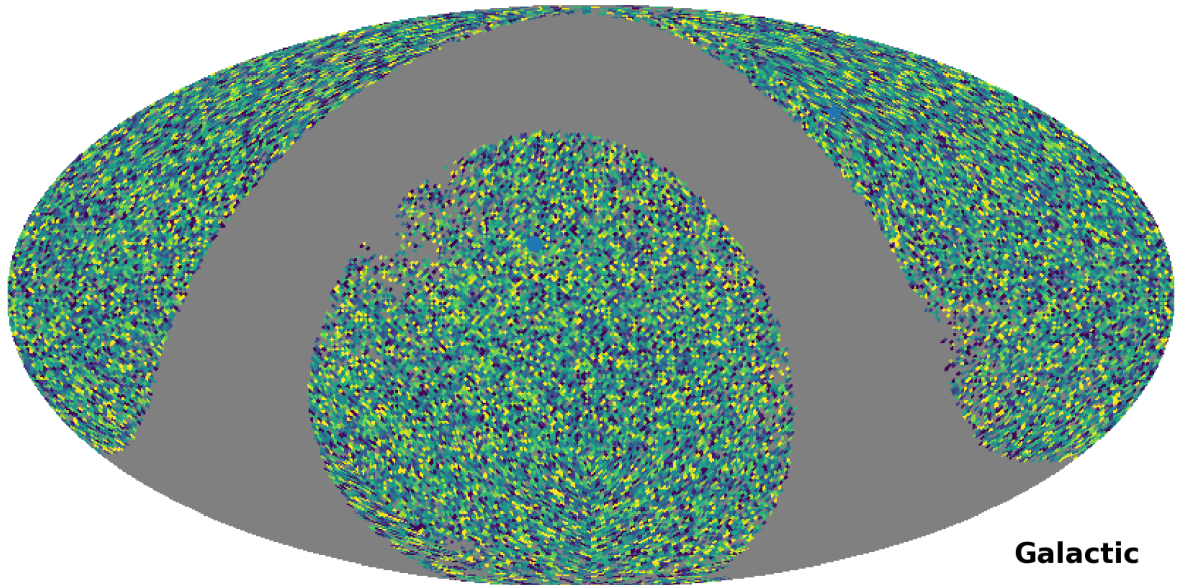




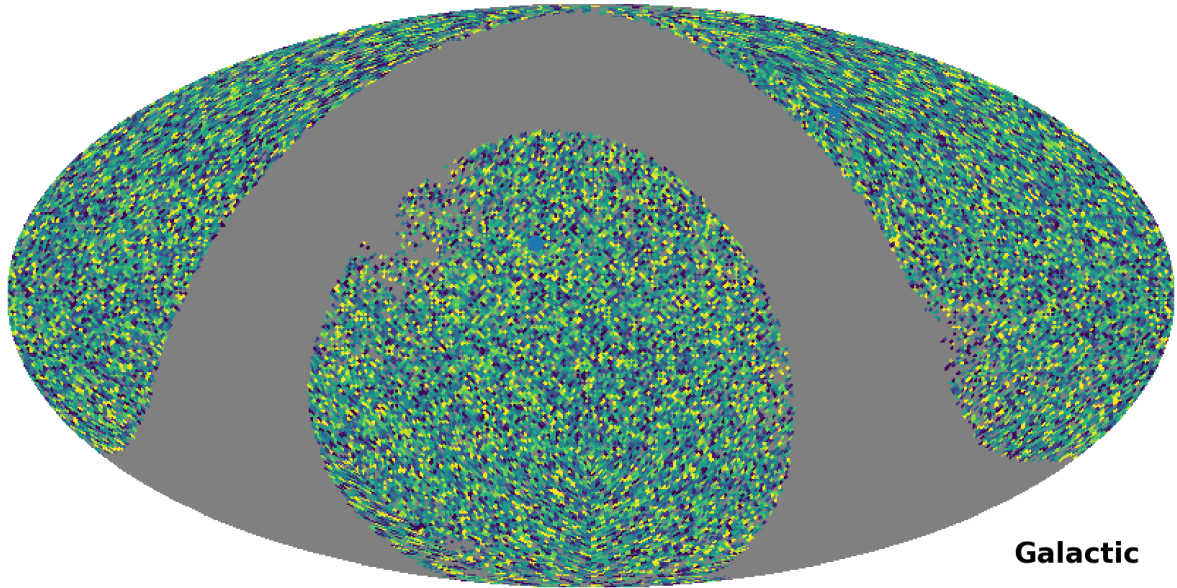
$0.75 \leq z < 1.00$  ( $|b| > 20^\circ$ ) residual  
 $z_{\text{resid}} = z - (a + bN)$ ,  $a=0.880$ ,  $b=1.98\text{e-}04$   
 dipole:  $\text{amp}=1.60\text{e-}03$ ,  $(l,b)=(203.3^\circ, -47.5^\circ)$



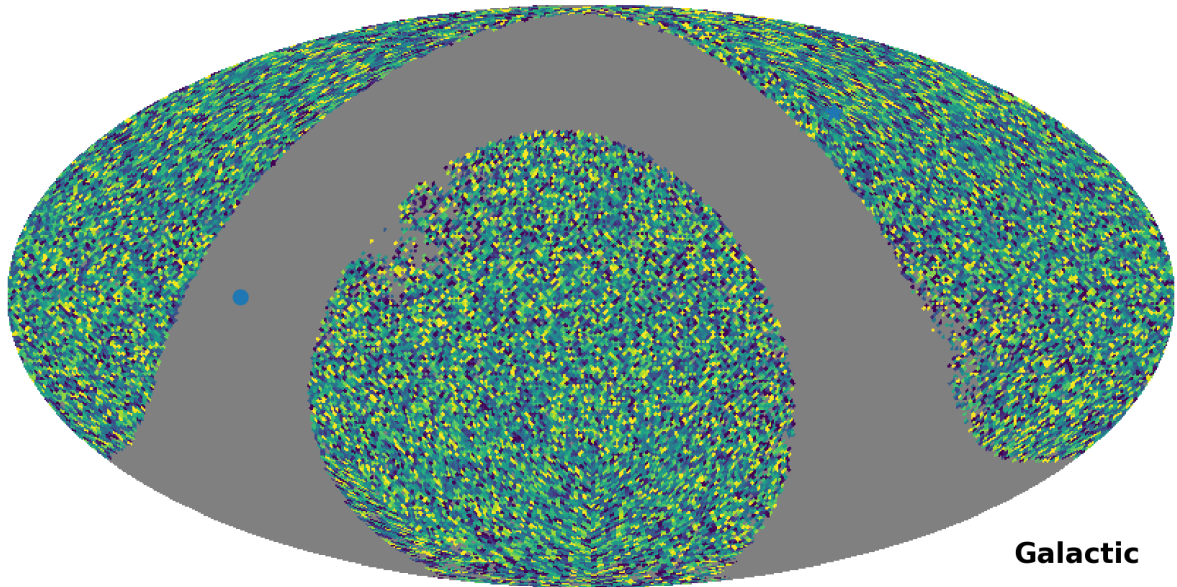
$1.00 \leq z < 1.50$  ( $|b| > 20^\circ$ )  
 $\langle z \rangle$  dipole:  $\text{amp}=1.85\text{e-}03$ ,  $(l,b)=(17.9^\circ, 13.3^\circ)$



$1.00 \leq z < 1.50$  ( $|b| > 20^\circ$ ) residual  
 $z_{\text{resid}} = z - (a + bN)$ ,  $a=1.259$ ,  $b=8.98\text{e-}05$   
 dipole:  $\text{amp}=1.85\text{e-}03$ ,  $(l,b)=(17.4^\circ, 13.1^\circ)$

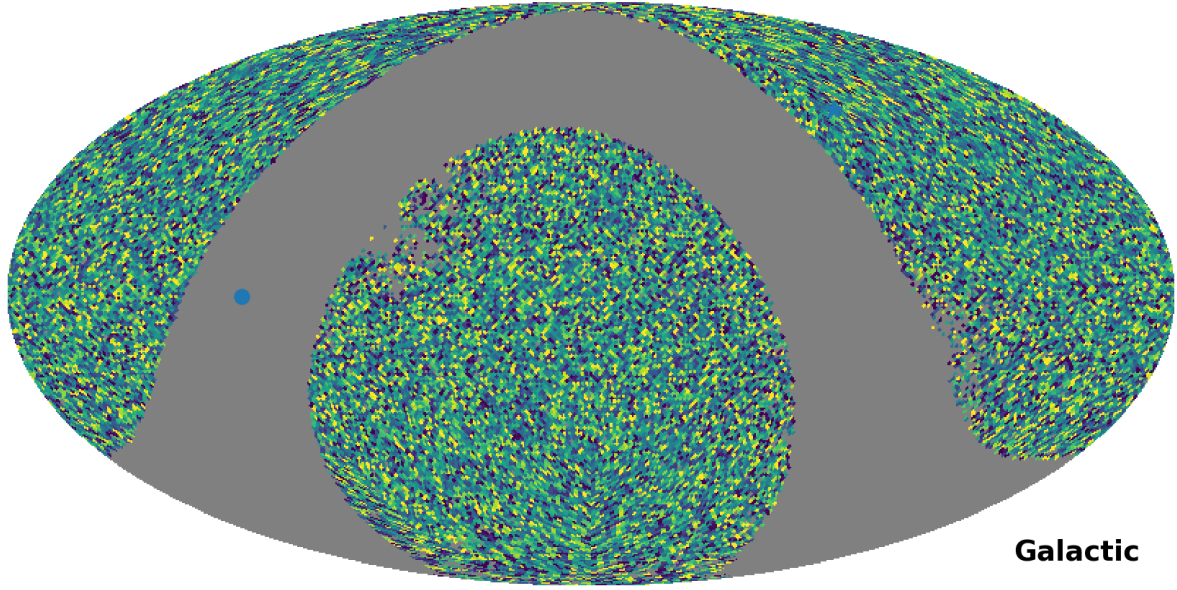


$1.50 \leq z < 2.50$  ( $|b| > 20^\circ$ )  
 $\langle z \rangle$  dipole:  $\text{amp}=3.48\text{e-}03$ ,  $(l,b)=(108.0^\circ, -0.2^\circ)$





$1.50 \leq z < 2.50$  ( $|b| > 20^\circ$ ) residual  
 $z_{\text{resid}} = z - (a + bN)$ ,  $a=1.926$ ,  $b=2.59\text{e-}04$   
 dipole:  $\text{amp}=3.39\text{e-}03$ ,  $(l,b)=(107.4^\circ, -0.8^\circ)$



| #      | bcut       | tag                | N_good       | amp        | l_deg     | b_deg      | sep_cmb |        |  |
|--------|------------|--------------------|--------------|------------|-----------|------------|---------|--------|--|
| 10.0   | full       |                    | 39147        | 1.190e-02  | 105.04    | 24.32      | 105.26  |        |  |
| 10.0   | z0p10_0p30 |                    | 232          | 6.269e-03  | 343.62    | 61.33      | 44.78   |        |  |
| 10.0   | z0p30_0p50 |                    | 3561         | 2.200e-03  | 81.08     | -11.21     | 143.13  |        |  |
| 10.0   | z0p50_0p75 |                    | 8510         | 2.027e-03  | 167.78    | -32.54     | 117.44  |        |  |
| 10.0   | z0p75_1p00 |                    | 14073        | 9.652e-04  | 151.43    | -30.84     | 126.97  |        |  |
| 10.0   | z1p00_1p50 |                    | 32040        | 1.941e-03  | 45.92     | 14.01      | 109.34  |        |  |
| 10.0   | z1p50_2p50 |                    | 35608        | 2.394e-03  | 82.06     | -3.06      | 135.03  |        |  |
| 20.0   | full       |                    | 32388        | 8.962e-03  | 110.95    | 18.51      | 109.25  |        |  |
| 20.0   | z0p10_0p30 |                    | 209          | 9.028e-03  | 325.48    | 51.12      | 38.83   |        |  |
| 20.0   | z0p30_0p50 |                    | 3237         | 1.829e-03  | 81.82     | -17.45     | 149.40  |        |  |
| 20.0   | z0p50_0p75 |                    | 7675         | 1.589e-03  | 155.87    | -17.47     | 114.94  |        |  |
| 20.0   | z0p75_1p00 |                    | 12687        | 1.311e-03  | 204.71    | -45.30     | 106.73  |        |  |
| 20.0   | z1p00_1p50 |                    | 28058        | 1.662e-03  | 34.87     | 12.30      | 105.64  |        |  |
| 20.0   | z1p50_2p50 |                    | 30800        | 2.953e-03  | 91.78     | -9.95      | 141.39  |        |  |
| 30.0   | full       |                    | 24778        | 5.187e-03  | 123.29    | 14.09      | 108.74  |        |  |
| 30.0   | z0p10_0p30 |                    | 165          | 5.275e-03  | 301.00    | 18.68      | 41.91   |        |  |
| 30.0   | z0p30_0p50 |                    | 2645         | 2.149e-03  | 95.39     | -30.99     | 160.90  |        |  |
| 30.0   | z0p50_0p75 |                    | 6291         | 1.236e-03  | 169.62    | -19.70     | 107.37  |        |  |
| 30.0   | z0p75_1p00 |                    | 10316        | 1.277e-03  | 217.18    | -39.81     | 97.13   |        |  |
| 30.0   | z1p00_1p50 |                    | 22073        | 2.321e-03  | 41.80     | 21.87      | 100.56  |        |  |
| 30.0   | z1p50_2p50 |                    | 24014        | 2.256e-03  | 95.58     | -24.80     | 155.06  |        |  |
|        |            |                    |              |            |           |            |         |        |  |
| #      | tag        | region             | N_use        | amp_full   | amp_jk    | d_amp_frac | l_full  | b_full |  |
| l_jk   | b_jk       | sep_full           | sep_cmb_full | sep_cmb_jk |           |            |         |        |  |
| full   |            | leave_N_hemisphere | 16120        | 8.962e-03  | 2.554e-02 | 1.850      |         |        |  |
| 110.95 | 18.51      | 60.77              | 12.13        | 48.66      | 109.25    | 116.42     |         |        |  |
| full   |            | leave_S_hemisphere | 16268        | 8.962e-03  | 2.061e-02 | 1.299      |         |        |  |
| 110.95 | 18.51      | 197.48             | 13.57        | 82.51      | 109.25    | 64.31      |         |        |  |
| full   |            | leave_Q1_0-90      | 23388        | 8.962e-03  | 1.493e-02 | 0.666      |         |        |  |
| 110.95 | 18.51      | 116.20             | 54.73        | 36.44      | 109.25    | 73.75      |         |        |  |

|            |        |                    |        |           |           |        |
|------------|--------|--------------------|--------|-----------|-----------|--------|
| full       |        | leave_Q2_90-180    | 25221  | 8.962e-03 | 1.426e-02 | 0.591  |
| 110.95     | 18.51  | 113.01_5.14        | 13.52  | 109.25    | 121.08    |        |
| full       |        | leave_Q3_180-270   | 23328  | 8.962e-03 | 1.081e-02 | 0.206  |
| 110.95     | 18.51  | 94.43_-31.24       | 52.22  | 109.25    | 161.46    |        |
| full       |        | leave_Q4_270-360   | 25227  | 8.962e-03 | 5.837e-03 | -0.349 |
| 110.95     | 18.51  | 122.56_38.93       | 22.77  | 109.25    | 86.56     |        |
| z0p10_0p30 |        | leave_N_hemisphere | 110    | 9.028e-03 | 1.424e-02 | 0.578  |
| 325.48     | 51.12  | 349.09_56.24       | 14.80  | 38.83     | 49.48     |        |
| z0p10_0p30 |        | leave_S_hemisphere | 99     | 9.028e-03 | 1.619e-02 | 0.793  |
| 325.48     | 51.12  | 283.53_-67.86      | 123.03 | 38.83     | 116.78    |        |
| z0p10_0p30 |        | leave_Q1_0-90      | 142    | 9.028e-03 | 1.082e-02 | 0.198  |
| 325.48     | 51.12  | 292.18_56.55       | 20.16  | 38.83     | 19.06     |        |
| z0p10_0p30 |        | leave_Q2_90-180    | 155    | 9.028e-03 | 9.675e-03 | 0.072  |
| 325.48     | 51.12  | 322.93_50.38       | 1.78   | 38.83     | 37.56     |        |
| z0p10_0p30 |        | leave_Q3_180-270   | 164    | 9.028e-03 | 8.761e-03 | -0.029 |
| 325.48     | 51.12  | 333.71_48.91       | 5.73   | 38.83     | 44.56     |        |
| z0p10_0p30 |        | leave_Q4_270-360   | 166    | 9.028e-03 | 7.932e-03 | -0.121 |
| 325.48     | 51.12  | 325.82_42.39       | 8.73   | 38.83     | 42.74     |        |
| z0p30_0p50 |        | leave_N_hemisphere | 1544   | 1.829e-03 | 4.105e-03 | 1.244  |
| 81.82      | -17.45 | 63.68_48.34        | 67.76  | 149.40    | 82.06     |        |
| z0p30_0p50 |        | leave_S_hemisphere | 1693   | 1.829e-03 | 1.439e-03 | -0.213 |
| 81.82      | -17.45 | 129.73_-30.97      | 45.36  | 149.40    | 141.53    |        |
| z0p30_0p50 |        | leave_Q1_0-90      | 2413   | 1.829e-03 | 2.029e-03 | 0.109  |
| 81.82      | -17.45 | 83.62_-6.55        | 11.04  | 149.40    | 138.55    |        |
| z0p30_0p50 |        | leave_Q2_90-180    | 2490   | 1.829e-03 | 1.771e-03 | -0.032 |
| 81.82      | -17.45 | 73.85_-32.19       | 16.41  | 149.40    | 162.42    |        |
| z0p30_0p50 |        | leave_Q3_180-270   | 2248   | 1.829e-03 | 7.757e-04 | -0.576 |
| 81.82      | -17.45 | 69.14_11.26        | 31.32  | 149.40    | 119.29    |        |
| z0p30_0p50 |        | leave_Q4_270-360   | 2560   | 1.829e-03 | 2.260e-03 | 0.236  |
| 81.82      | -17.45 | 94.95_-26.96       | 15.41  | 149.40    | 157.29    |        |
| z0p50_0p75 |        | leave_N_hemisphere | 3884   | 1.589e-03 | 3.605e-03 | 1.269  |
| 155.87     | -17.47 | 167.98_71.82       | 89.67  | 114.94    | 46.83     |        |
| z0p50_0p75 |        | leave_S_hemisphere | 3791   | 1.589e-03 | 4.644e-03 | 1.923  |
| 155.87     | -17.47 | 148.77_-67.12      | 49.86  | 114.94    | 142.71    |        |
| z0p50_0p75 |        | leave_Q1_0-90      | 5509   | 1.589e-03 | 1.975e-03 | 0.243  |
| 155.87     | -17.47 | 122.98_-11.89      | 32.26  | 114.94    | 131.46    |        |
| z0p50_0p75 |        | leave_Q2_90-180    | 5938   | 1.589e-03 | 1.051e-03 | -0.339 |
| 155.87     | -17.47 | 238.55_-45.07      | 72.64  | 114.94    | 95.70     |        |
| z0p50_0p75 |        | leave_Q3_180-270   | 5475   | 1.589e-03 | 1.870e-03 | 0.177  |
| 155.87     | -17.47 | 190.19_0.29        | 38.16  | 114.94    | 79.03     |        |
| z0p50_0p75 |        | leave_Q4_270-360   | 6103   | 1.589e-03 | 2.490e-03 | 0.567  |
| 155.87     | -17.47 | 151.36_-20.88      | 5.46   | 114.94    | 120.37    |        |
| z0p75_1p00 |        | leave_N_hemisphere | 6242   | 1.311e-03 | 1.037e-03 | -0.209 |
| 204.71     | -45.30 | 306.26_75.94       | 136.36 | 106.73    | 32.74     |        |
| z0p75_1p00 |        | leave_S_hemisphere | 6445   | 1.311e-03 | 2.330e-03 | 0.778  |
| 204.71     | -45.30 | 199.85_20.97       | 66.42  | 106.73    | 57.43     |        |
| z0p75_1p00 |        | leave_Q1_0-90      | 9269   | 1.311e-03 | 5.333e-04 | -0.593 |
| 204.71     | -45.30 | 133.33_-47.11      | 47.65  | 106.73    | 147.28    |        |
| z0p75_1p00 |        | leave_Q2_90-180    | 9679   | 1.311e-03 | 8.897e-04 | -0.321 |
| 204.71     | -45.30 | 204.40_-28.23      | 17.08  | 106.73    | 93.05     |        |
| z0p75_1p00 |        | leave_Q3_180-270   | 9109   | 1.311e-03 | 2.092e-03 | 0.596  |
| 204.71     | -45.30 | 220.63_-51.84      | 12.36  | 106.73    | 106.49    |        |
| z0p75_1p00 |        | leave_Q4_270-360   | 10004  | 1.311e-03 | 1.892e-03 | 0.444  |
| 204.71     | -45.30 | 180.23_-41.07      | 18.27  | 106.73    | 115.69    |        |
| z1p00_1p50 |        | leave_N_hemisphere | 13918  | 1.662e-03 | 1.883e-03 | 0.133  |
| 34.87      | 12.30  | 85.50_10.34        | 49.61  | 105.64    | 121.64    |        |
| z1p00_1p50 |        | leave_S_hemisphere | 14140  | 1.662e-03 | 2.566e-03 | 0.543  |
| 34.87      | 12.30  | 1.75_0.83          | 34.79  | 105.64    | 94.56     |        |

|             |                    |       |           |           |        |
|-------------|--------------------|-------|-----------|-----------|--------|
| z1p00_1p50  | leave_Q1_0-90      | 20452 | 1.662e-03 | 9.541e-04 | -0.426 |
| 34.87_12.30 | 289.78_29.68       | 96.64 | 105.64    | 26.94     |        |
| z1p00_1p50  | leave_Q2_90-180    | 21655 | 1.662e-03 | 2.322e-03 | 0.397  |
| 34.87_12.30 | 72.48_21.32        | 37.01 | 105.64    | 109.91    |        |
| z1p00_1p50  | leave_Q3_180-270   | 20105 | 1.662e-03 | 2.591e-03 | 0.559  |
| 34.87_12.30 | 42.30_8.45         | 8.26  | 105.64    | 112.64    |        |
| z1p00_1p50  | leave_Q4_270-360   | 21962 | 1.662e-03 | 2.029e-03 | 0.221  |
| 34.87_12.30 | 13.28_-2.44        | 26.03 | 105.64    | 104.62    |        |
| z1p50_2p50  | leave_N_hemisphere | 15329 | 2.953e-03 | 5.144e-03 | 0.742  |
| 91.78_-9.95 | 87.02_4.30         | 15.02 | 141.39    | 127.63    |        |
| z1p50_2p50  | leave_S_hemisphere | 15471 | 2.953e-03 | 1.302e-03 | -0.559 |
| 91.78_-9.95 | 131.13_46.50       | 66.49 | 141.39    | 76.96     |        |
| z1p50_2p50  | leave_Q1_0-90      | 22352 | 2.953e-03 | 2.419e-03 | -0.181 |
| 91.78_-9.95 | 109.66_12.96       | 28.98 | 141.39    | 114.90    |        |
| z1p50_2p50  | leave_Q2_90-180    | 23886 | 2.953e-03 | 4.749e-03 | 0.608  |
| 91.78_-9.95 | 100.13_3.21        | 15.57 | 141.39    | 126.88    |        |
| z1p50_2p50  | leave_Q3_180-270   | 22190 | 2.953e-03 | 3.149e-03 | 0.066  |
| 91.78_-9.95 | 81.61_-31.35       | 23.39 | 141.39    | 163.25    |        |
| z1p50_2p50  | leave_Q4_270-360   | 23972 | 2.953e-03 | 2.559e-03 | -0.134 |
| 91.78_-9.95 | 88.01_-21.11       | 11.74 | 141.39    | 152.92    |        |